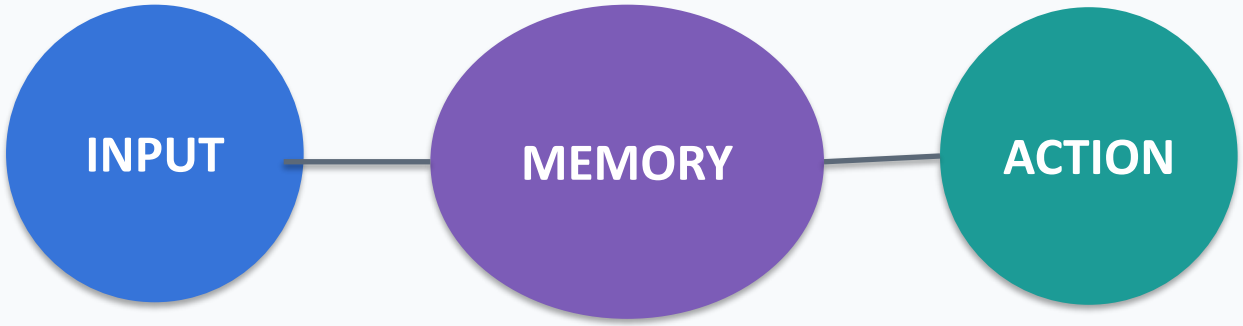


Core idea

Good interfaces work with human memory—not against it.

Course focus

Memory models • limitations • recognition • design principles • examples

Human Memory in HCI

How users remember, forget, recognize, learn and interact with digital systems

Learning Objectives

- Define human memory and explain why it matters in HCI.
- Differentiate sensory memory, working/short-term memory and long-term memory.
- Explain recognition vs. recall and why recognition is usually easier.
- Identify common memory limitations that cause usability problems.
- Apply memory principles to websites, mobile apps, ATMs, dashboards and games.
- Design interfaces that reduce unnecessary memory load.

Why Does Memory Matter in HCI?

Users are not computers

People have limited attention and working memory. Interfaces must not force users to remember too much.

Everyday example

A user moves between pages and forgets a 6-digit code or where a setting was located.

HCI goal

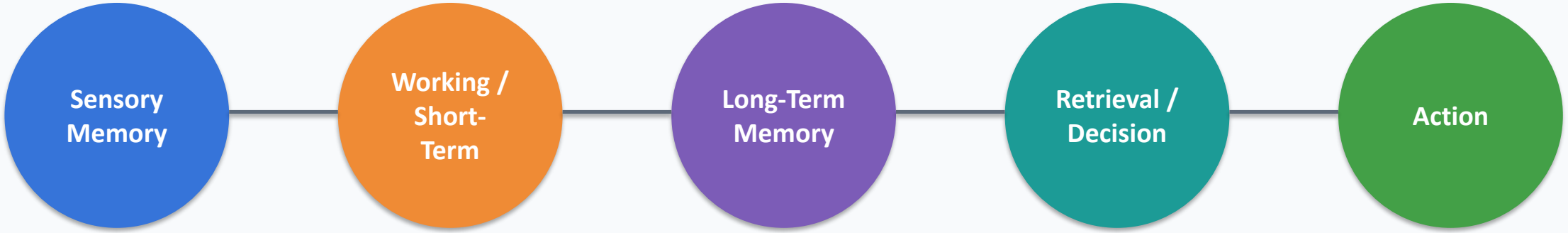
Keep important information visible, recognizable and easy to retrieve.

Design principle

If the system can remember it for the user, the user should not have to.

Human Memory: The Big Picture

A simplified information-processing view



HCI connection

A user sees information → attends to it → holds it temporarily → connects it to knowledge → chooses an action.

1. Sensory Memory

- Very brief storage of information received through the senses.
- Visual information can persist for a fraction of a second; auditory information can last a few seconds.
- Most sensory information disappears unless attention selects it.
- Design implication: visual hierarchy and clear feedback help users notice the right thing.

Example: Notification

A short toast message appears: “Saved successfully.” It must be noticed quickly.

Good design

Use readable contrast, sensible duration and a clear location for transient feedback.

Bad design

A tiny, low-contrast message disappears before the user can see it.

2. Working Memory (Short-Term Memory)

- Temporary mental workspace used to hold and manipulate information.
- Capacity is limited; exact capacity varies with task, expertise and chunking.
- Information is easily disrupted by distraction or competing tasks.
- Users struggle when an interface requires them to remember several values while navigating elsewhere.

Unchunked

9 2 4 7 1 6 3 8

Chunked

924 – 716 – 38

HCI lesson

Group related information instead of presenting many isolated items.

Working Memory in a Real Interface

Bad checkout flow

1. Remember product price
2. Go to another screen
3. Remember discount code
4. Return to checkout
5. Re-enter shipping details
6. Calculate total mentally

Better checkout flow

✓ Keep cart summary visible
✓ Auto-apply discounts
✓ Save entered details
✓ Show subtotal + shipping + total
✓ Provide Back/Next without losing data

3. Long-Term Memory

- Stores knowledge, skills, experiences and associations over longer periods.
- Includes facts (semantic memory), events (episodic memory) and learned skills/procedures.
- Prior experience strongly affects how users interpret an interface.
- Familiar patterns reduce learning time.

Semantic

"A shopping cart contains items I intend to buy."

Episodic

"I used this app yesterday to submit an assignment."

Procedural

Knowing how to swipe, drag, scroll or type.

Recognition vs. Recall

Recognition

User identifies something when it is presented.

Example: Selecting “Settings” from a visible menu.

Recall

User must retrieve information without seeing it.

Example: Typing the exact name of a hidden command.

HCI rule of thumb

Prefer recognition over recall whenever practical: show options, labels, examples, recent items and suggestions.

Recognition in Everyday Interfaces

ATM

Show account options and amounts

Google-style search

Autocomplete reduces typing/recall

Phone keyboard

Predictive suggestions

File manager

Visible folders and icons

E-commerce

Recent searches/products

Maps

Saved places and suggestions

Chunking: Making Information Easier to Remember

Raw information

0315123456789
A long sequence is difficult to hold and compare.

Chunked information

03 – 151 – 234 – 56789
Groups create meaningful units.

HCI applications

Phone numbers • dates • credit-card fields • menus • navigation steps

Design benefit

Less mental effort → fewer errors → faster completion

Mental Models & Memory

- A mental model is the user's internal expectation of how a system works.
- People use previous experiences to predict what will happen next.
- Familiar conventions reduce the need to learn new commands.
- Breaking conventions without a strong reason can increase errors and confusion.

Familiar

Trash/bin icon → delete

Familiar

Magnifying glass → search

Unexpected

A floppy-disk icon behaving as “Delete”

Forgetting: Why Users Make Mistakes

Decay / weak memory

Information that is not used may become harder to retrieve.

Interference

New information can compete with older information.

Context change

Information is harder to retrieve when the situation changes.

HCI response

Use visible state, reminders, history, defaults, confirmations, undo, autosave and persistent labels.

Prospective Memory: Remembering to Do Something

- Prospective memory means remembering a future intention.
- Digital systems can support it with reminders, notifications and clear task states.
- Example: “Submit assignment before 11:59 PM.”
- Poorly timed notifications can become noise and be ignored.

Helpful “Assignment due tomorrow” + clear action button

Unhelpful

Repeated irrelevant notifications

Design

Let users control timing, frequency and priority.

Memory Load: Common HCI Problems

Too many steps

User must remember previous choices

Hidden commands

User must recall where a feature is

Inconsistent labels

Same function has different names

Form resets

User must re-enter information

No history

User cannot retrieve previous work

No feedback

User cannot remember whether action succeeded

Design Principles: Reduce Memory Load

- Make important information visible at the point of use.
- Prefer recognition over recall.
- Use familiar conventions and consistent terminology.
- Chunk long information into meaningful groups.
- Provide defaults, autocomplete and suggestions.
- Preserve user input and progress.
- Provide history, recent items and undo.
- Give clear feedback after actions.

Case Study: University Student Portal

Problem

A student logs into the portal to register courses.

The system makes the student remember:

- course codes
- section numbers
- fee amount
- previous selections
- where to find timetable

Result: errors, backtracking and frustration.

Memory-friendly redesign

✓ Show course names + codes together
✓ Keep selected courses visible
✓ Show timetable preview
✓ Save progress automatically
✓ Provide search + filters
✓ Use consistent navigation
✓ Confirm successful registration

Case Study: Mobile Banking

High memory load

User must remember:

- account number
- beneficiary details
- OTP timing
- transfer amount
- previous screen

Better design

- Saved beneficiaries
- Visible account name
- Amount summary
- OTP countdown
- Review screen
- Confirmation receipt

Result

Less recall

↓

Less cognitive effort

↓

Fewer mistakes

↓

Higher confidence

Human Memory + HCI: Quick Comparison

Memory issue	Poor interface	Memory-friendly interface
Recall	Type exact command	Show command / menu
Recognition	Find familiar option	Visible labels + icons
Working memory	Remember previous page	Persistent summary
Long-term memory	Use learned convention	Follow familiar patterns
Prospective memory	Remember future task	Reminder + task status
Forgetting	Re-enter lost data	Autosave + history

Class Activity: Memory Detective 🔎

Task

Choose one app you use every day.

1. Find one place where the app makes you remember something.
2. Is it recall or recognition?
3. What memory type is involved?
4. Redesign that part to reduce memory load.
5. Sketch your improved screen in 5 minutes.

Examples to investigate

Instagram / WhatsApp / YouTube / Google Maps / Banking app / University portal / Food delivery app

Bonus: Find one example of chunking, autocomplete, history, saved items or a familiar icon.

Exam / Discussion Questions

- Why is recognition generally easier than recall? Give one HCI example.
- Differentiate sensory, working and long-term memory.
- How can chunking improve interface usability?
- Identify three ways a student portal can reduce memory load.
- Explain how mental models are connected to previous experience.
- Design one memory-friendly screen for an ATM, mobile app or university portal.

Key Takeaways

1. Memory is limited

Do not overload the user.

2. Recognition wins

Show choices instead of demanding recall.

3. Use familiar patterns

Past experience is a design resource.

4. Support retrieval

History, reminders and visible state help.

5. Reduce mental effort

Chunk, simplify and keep context visible.

6. Design for humans

The interface should carry part of the cognitive load.

Final message

Good HCI does not test the user's memory—it supports it.